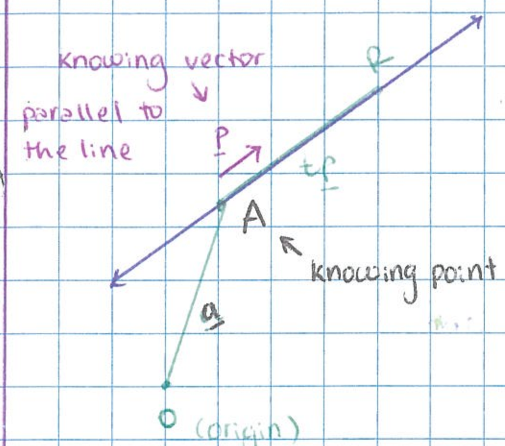
PURE

VECTORS

THE VECTOR EQUATION OF A LINE



Given a point A on the line and a vector $\underline{p}$ parallel to line, any other point R on the line has position vector

$$\underline{r} = \underline{a} + t\underline{p}$$

$t \rightarrow \lambda, \mu, s$ (constant)

$\underline{a}$: point on the line

$\underline{p}$: direction

E.g. Give the vector equation of the line through the point $(6, 1, -2)$ parallel to the vector $\begin{pmatrix} 3 \\ -2 \\ 0 \end{pmatrix}$

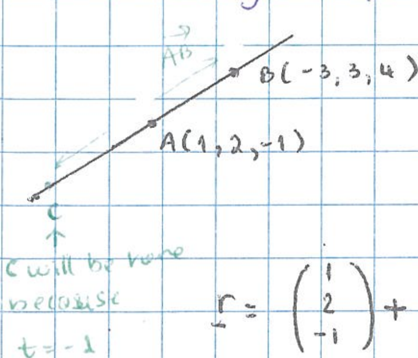
Ans

$$\underline{r} = \underline{a} + t\underline{p}$$

$$\underline{r} = \begin{pmatrix} 6 \\ 1 \\ -2 \end{pmatrix} + t\begin{pmatrix} 3 \\ -2 \\ 0 \end{pmatrix}$$

E.g (a) Find the vector equation of the line through $A(1, 2, -1)$ and $B(-3, 3, 4)$

(b) use your equation to determine if $C(5, 1, -6)$ lies on this line.



(a) $\underline{r} = \underline{a} + t\underline{p}$

$$\underline{p} = \vec{AB} = \begin{pmatrix} -3 \\ 3 \\ 4 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} = \begin{pmatrix} -4 \\ 1 \\ 5 \end{pmatrix}$$

$$\underline{r} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} + t\begin{pmatrix} -4 \\ 1 \\ 5 \end{pmatrix}$$

or

$$= \begin{pmatrix} -3 \\ 3 \\ 4 \end{pmatrix} + t\begin{pmatrix} -4 \\ 1 \\ 5 \end{pmatrix}$$

(b) solve $\begin{pmatrix} 5 \\ 1 \\ -6 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} + t\begin{pmatrix} -4 \\ 1 \\ 5 \end{pmatrix}$

t must work for all 3 components.

x : $5 = 1 - 4t \Rightarrow t = -1$

y : $1 = 2 + t \Rightarrow t = -1$

z : $-6 = -1 + 5t \Rightarrow t = -1$

Yes, C is on the line (when $t = -1$)

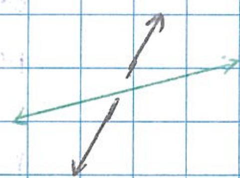
TWO LINES IN 3D

4 scenarios :

1. Single point of intersection

2. No intersection and parallel

3. No intersection and skew (not parallel)



4. Always intersecting - ie the same line

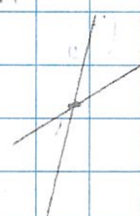
SOLVING SIMULTANOUS VECTOR EQUATIONS

e.g Do lines 1 and 2 intersect ?
If so, where ?

direction

line 1 $\quad r = \begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ -1 \\ 2 \end{pmatrix}$

line 2 $\quad r = \begin{pmatrix} 0 \\ -2 \\ 8 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$



direction multiple to each other
line 1 // line 2
$\lambda \begin{pmatrix} -2 \\ -1 \\ 2 \end{pmatrix} \quad \mu \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}$

Solve : $\begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ -1 \\ 2 \end{pmatrix} = \begin{pmatrix} 0 \\ -2 \\ 8 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$

$$\left.\begin{aligned} x &: 1 - 2\lambda = 0 + \mu \\ y &: 3 - \lambda = -2 - \mu \\ z &: 1 + 2\lambda = 8 + \mu \end{aligned}\right\}$$ can we solve all 3 equations for consistent λ and μ

Take 2 equations - solve for μ, λ
then subst. into 3^{rd} equation

$$\begin{aligned} 1 - 2\lambda &= 0 + \mu \\ 3 - \lambda &= -2 - \mu \\ \hline 4 - 3\lambda &= -2 \\ 3\lambda &= 6 \\ \lambda &= 2 \end{aligned}$$

$$\begin{aligned} 3 - 2 &= -2 - \mu \\ \mu &= -3 \end{aligned}$$

Subst. into $1 + 2\lambda = 8 + \mu$

Does $1 + 2(2) = 8 + (-3)$?

$5 = 5$

If $5 \neq 1$, don't intesect, so decide whether they are // or skew

$\therefore$ ~~so,~~ yes. There is consistent solution

So, lines intersect when

$\lambda = 2, \mu = -3$

i) sub $\lambda = 2$ into line 1

$$r = \begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix} + 2\begin{pmatrix} -2 \\ -1 \\ 2 \end{pmatrix} = \begin{pmatrix} -3 \\ 1 \\ 5 \end{pmatrix}$$

$\therefore$ point of intersection

$(-3, 1, 5)$

19.6.20

Example : Line 1 : $r = \begin{pmatrix} 1 \\ 1 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ -3 \end{pmatrix}$

Line 2 : $r = \begin{pmatrix} -3 \\ -1 \\ 10 \end{pmatrix} + \mu \begin{pmatrix} -2 \\ -1 \\ 3 \end{pmatrix}$

Do the lines intersect ? if so, where ?

$x: 1 + 2\lambda = -3 - 2\mu$ (1)

$y: 1 + \lambda = -1 - \mu$ (2)

$z: 4 - 3\lambda = 10 + 3\mu$ (3)

$$\begin{aligned} 2 \times (2): \quad 2 + 2\lambda &= -2 - 2\mu \\ - \quad 1 + 2\lambda &= -3 - 2\mu \\ \hline 1 + 0 &= 1 + 0 \\ 1 &= 1 \end{aligned}$$

Equations are consistent for any values of λ or μ

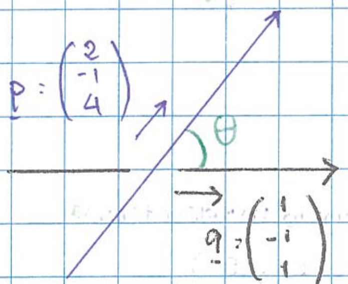
$$\begin{aligned} 3 \times (2): \quad 3 + 3\lambda &= -3 - 3\mu \\ + \quad 4 - 3\lambda &= 10 + 3\mu \\ \hline 7 &= 7 \end{aligned}$$

So, the 2 lines are in fact the same line.

FINDING THE ANGLE BETWEEN 2 LINES

Line 1 : $\underline{r} = \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -1 \\ 4 \end{pmatrix}$

Line 2 : $\underline{r} = \begin{pmatrix} 2 \\ 4 \\ 4 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$



$\underline{p} = \begin{pmatrix} 2 \\ -1 \\ 4 \end{pmatrix}$ $\quad$ $\underline{q} = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$

$$\cos\theta = \frac{\underline{p} \times \underline{q}}{|p||q|}$$

in vector, always work in degree (1 d.p)

$$\cos\theta = \frac{\begin{pmatrix} 2 \\ -1 \\ 4 \end{pmatrix} \times \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}}{\sqrt{2^2+(-1)^2+4^2}\cdot\sqrt{1^2+(-1)^2+1^2}} = \frac{2+1+4}{\sqrt{21} \times \sqrt{3}}$$

$$\cos\theta = \frac{7}{\sqrt{63}}$$

$$\theta = \cos^{-1}\left(\frac{7}{\sqrt{63}}\right)$$

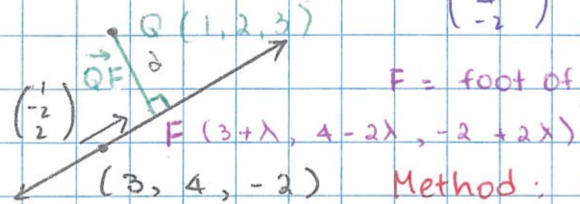
$$\theta = 28.1255\ldots = 28.1 \ (1\,dp)$$

NOTE : If the calculation gives an obtuse angle (90 - 180) then the acute angle is found from 180° - obtuse angle

FINDING THE SHORTEST DISTANCE OF A POINT FROM A LINE

E.g Find the shortest distance from the point $Q(1,2,3)$ from the line $r = \begin{pmatrix} 3 \\ 4 \\ -2 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -2 \\ 2 \end{pmatrix}$

{ Shortest distance; perpendicular }



F = foot of the perpendicular

Method: Find coordinates of F then find distance $|\vec{QF}|$

$$r = \begin{pmatrix} 3+\lambda \\ 4-2\lambda \\ -2+2\lambda \end{pmatrix} \begin{array}{l} \leftarrow x \\ \leftarrow y \\ \leftarrow z \end{array}$$

So, any point on line has coordinates
$(3+\lambda,\ 4-2\lambda,\ -2+2\lambda)$

- Find vector $\vec{QF}$

$$\vec{QF} = \begin{pmatrix} 3+\lambda \\ 4-2\lambda \\ -2+2\lambda \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 2+\lambda \\ 2-2\lambda \\ -5+2\lambda \end{pmatrix}$$

Since $\vec{QF}$ is perpendicular to line,

$$\begin{pmatrix} 2+\lambda \\ 2-2\lambda \\ -5+2\lambda \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -2 \\ 2 \end{pmatrix} = 0$$

$$1(2+\lambda) - 2(2-2\lambda) + 2(-5+2\lambda) = 0$$

$$2 + \lambda - 4 + 4\lambda - 10 + 4\lambda = 0$$

$$9\lambda = 12$$

$$\lambda = \frac{12}{9} = \frac{4}{3}$$

So, $F = \left(3 + \frac{4}{3},\ 4 - 2\left(\frac{4}{3}\right),\ -2 + 2\left(\frac{4}{3}\right)\right)$

$$= \left(\frac{13}{3},\ \frac{4}{3},\ \frac{2}{3}\right)$$

and $|\vec{QF}|$

$$= \sqrt{\left(\frac{13}{3}-1\right)^2 + \left(\frac{4}{3}-2\right)^2 + \left(\frac{2}{3}-3\right)^2}$$

$$= \sqrt{\frac{100}{9} + \frac{4}{9} + \frac{49}{9}}$$

$$= \sqrt{17}$$